Qi Le

AI / ML Research Intern

Minneapolis, MN | le000288@umn.edu | +1 (518) 961-5098
qi-le1.github.io | github.com/Qi-Le1 | Google Scholar | LinkedIn

Ph.D. student researching efficient and trustworthy language models. First-author work on dynamic LLM pruning and federated learning; experience in model compression, adaptation, alignment, and GPU-based evaluation.

Education

University of Minnesota, Twin Cities — Ph.D. in Computer Science — Sep 2022 – Present

University of Minnesota, Twin Cities — M.S. in Data Science — Aug 2020 – May 2022

Rensselaer Polytechnic Institute — B.S. in Chemical Engineering; CS minor — Aug 2016 – May 2020

Technical Skills

Research stack: Python, PyTorch, Hugging Face Transformers / Datasets, NumPy; GPU inference benchmarking.

Methods: LLM pruning and merging, KV-cache compression, PEFT, alignment, federated learning.

Research Experience

Research Assistant, University of Minnesota, Twin Cities — May 2021 – Present

Advised by Prof. Ali Anwar, Prof. Jie Ding, and Dr. Enmao Diao.

Probe Pruning: Dynamic LLM Inference — First author | ICLR 2025

- Developed batch-adaptive structured pruning using lightweight model probes and historical activations to select attention heads and MLP channels, without fine-tuning; evaluated on LLaMA-2/3 and OPT.
- Achieved **1.46× attention-block and 1.30× MLP-block speedups** at 40% pruning on LLaMA-2-7B / WikiText2 (NVIDIA A100). Released open-source code.

Model Merging and KV-Cache Compression — First author | ICLR 2027 submissions

- Researching model merging through iterative parameter replacement (**OBMerge**) and KV-cache eviction with output-projected, OBS-inspired compensation (**ProjCache**).

Structured Pruning for Language and Reasoning Models — Coauthor | ACL / DAC 2026

- Contributed to GISP, an iterative, task-aligned global pruning framework. At 50% pruning, obtained **1.70× prefill throughput and 45% lower inference memory** on LLaMA-2-7B (A6000, batch 1). **ACL Outstanding Paper.**
- Contributed to RESP, which regenerates reasoning traces during progressive pruning; retained **81.27% GSM8K accuracy at 40% pruning** on Qwen3-8B, versus 14.40% for GISP in the same evaluation.

LLM Alignment and Content Safety — Coauthor | ICLR / NAACL 2025

- Contributed to MAP, using constrained optimization to align language models with user-defined human-value targets; evaluated decoding- and PPO-based alignment across six reward dimensions. **ICLR Oral.**
- Contributed to AID, combining eight safety detectors with input-dependent weights; achieved a **0.12 median absolute AUC gain** over the best individual detector across benchmark detection tasks.

Federated Learning and Efficient Model Adaptation — First / second author | 2024

- Developed DynamicFL to allocate client communication frequencies under resource budgets. Improved CIFAR-10 accuracy from **44.4% to 49.8%** over random client selection at a matched communication schedule (CNN, one class per client). **Best Student Paper.**
- Contributed to ColA, offloading adapter-gradient computation to separate devices. Llama-2 adaptation used **42.8 GB on the primary GPU** at batch 32, while the corresponding LoRA setup exceeded 48 GB (Dolly, all projections; extra offload-device memory).

Publications

OBMerge: Optimal Brain Merge via Iterative Parameter Replacement
Q. Le, E. Diao, Z. Wang, X. Wang, L. Yang, J. Ding, A. Anwar | *Submitted to ICLR 2027*

ProjCache: Output-Projected KV Cache Eviction with OBS-Inspired Compensation
Q. Le, E. Diao, Z. Wang, X. Wang, L. Yang, J. Ding, A. Anwar | *Submitted to ICLR 2027*

Think Before You Prune: Self-Reflective Structured Pruning for Reasoning Language Models
Z. Wang, E. Diao, **Q. Le**, et al. | **DAC 2026**

From Local to Global: Revisiting Structured Pruning Paradigms for Large Language Models
Z. Wang, E. Diao, **Q. Le**, et al. | **ACL 2026 — Outstanding Paper**

AHE: Adaptive Homomorphic Encryption for Customizable Privacy in Heterogeneous Federated Learning
J. Tang, X. Wang, **Q. Le**, et al. | **IEEE INFOCOM 2026**

Beyond Expectations: Quantile-Guided Alignment for Risk-Calibrated Language Models
X. Wang, J. Du, A. A. Khan, **Q. Le**, et al. | **NeurIPS 2025 — Spotlight**

Probe Pruning: Accelerating LLMs through Dynamic Pruning via Model-Probing
Q. Le, E. Diao, Z. Wang, et al. | **ICLR 2025**

MAP: Multi-Human-Value Alignment Palette
X. Wang, **Q. Le**, A. Ahmed, et al. | **ICLR 2025 — Oral**

AID: Adaptive Integration of Detectors for Safe AI with Language Models
X. Wang, E. Diao, **Q. Le**, et al. | **NAACL 2025**

IP-FL: Incentive-Driven Personalization in Federated Learning
A. F. Khan, X. Wang, **Q. Le**, et al. | **IEEE IPDPS 2025**

DynamicFL: Federated Learning with Dynamic Communication Resource Allocation
Q. Le, E. Diao, X. Wang, et al. | **IEEE BigData 2024 — Best Student Paper**

ICL: An Incentivized Collaborative Learning Framework
X. Wang, **Q. Le**, A. F. Khan, et al. | **IEEE BigData 2024**

ColA: Collaborative Adaptation with Gradient Learning
E. Diao, **Q. Le**, S. Wu, et al. | *arXiv:2404.13844, preprint, 2024*

Personalized Federated Recommender Systems with Private and Partially Federated AutoEncoders
Q. Le, E. Diao, X. Wang, et al. | **Asilomar 2022**

Honors and Awards

Outstanding Paper Award, ACL 2026

Machine Learning Systems Fellowship, Amazon 2025–2026

Best Student Paper Award, IEEE BigData 2024

Teaching and Professional Service

Teaching Assistant, University of Minnesota Mar 2021 – Present
Business Analytics; Algorithms and Data Structures; Introduction to Algorithms and Data Structures.

Reviewer: ICLR 2027; NeurIPS 2025; Neural Processing Letters and IEEE BigData 2023; Asilomar 2022.

Co-reviewer (assisting faculty): USENIX ATC; USENIX FAST; IEEE Transactions on Parallel and Distributed Systems (TPDS).